# Linear functions — Part 1 answer key

| # | ID | Answer | Difficulty |
|---|---|---|---|
| 1.1 | fa992b50-1873-43ba-8b60-8c6d6bdc7bcc | 9 | Easy |
| 1.2 | c31256e3-e43c-46a7-998b-9e8511243e7b | C | Easy |
| 1.3 | df1e58e8-b763-42a1-a3e2-b70a4bac09d5 | D | Easy |
| 1.4 | 6aba0416-d25e-4a28-b655-c5623b3b71c0 | 2 | Easy |
| 1.5 | b347b50a-ec16-402a-b7e5-8ec5a188bf8f | C | Easy |
| 1.6 | 2050beca-5f4f-41a5-92ca-3fcf29a6cc77 | D | Easy |
| 1.7 | 821016f4-912a-4737-9916-2b5d5ea434a3 | C | Easy |
| 1.8 | 029367-DC | A | Easy |
| 1.9 | 6256e60f-a07b-4308-a386-94ecb7e20908 | A | Easy |
| 1.10 | 6ef925eb-e107-4c5c-a9ee-a6f89bf6d14e | 20 | Easy |
| 1.11 | 029386-DC | B | Easy |
| 1.12 | 84994068-0631-49ef-ae80-b174172efcd3 | B | Easy |
| 1.13 | f1332d3b-a308-4640-95e0-cae936826d8d | C | Easy |
| 1.14 | f7cecf09-36cf-4934-b397-765b7be1e77c | B | Easy |
| 1.15 | 020326-DC | A | Easy |
| 1.16 | b5527b99-0bdb-46d2-9513-73a5fae75276 | B | Easy |
| 1.17 | b86303fd-698f-45e4-ba2a-97ad8f15a2d6 | 224 | Easy |
| 1.18 | 2825ecca-ede4-4a29-bbb0-928be95b590b | 50 | Easy |
| 1.19 | 61063a9c-5dc4-47b2-858c-6c3052f4e804 | A | Easy |
| 1.20 | 08176-DC | C | Easy |
| 1.21 | 68357233-11ec-44c6-882e-eb8bb84450eb | 1 | Easy |
| 1.22 | 3e4d926d-83f1-4d65-87c3-c5889ee7bfa4 | 69 | Easy |

| | | | |
|---|---|---|---|
| 1.23 | 9db3fbf3-8d74-4521-aab7-7ad6bfc05c13 | B | Easy |
| 1.24 | d06b80b0-bba8-4218-bf79-f285777b8fbc | A | Easy |
| 1.25 | e35017e8-7f1b-42e1-a7bd-e47cb7628320 | B | Easy |
| 1.26 | 4c2cb31a-9ec5-40cc-863d-4baac298c63b | C | Easy |
| 1.27 | 9dc23c3a-e1d2-402a-9e2a-4085bbba4933 | A | Easy |
| 1.28 | 07aee0d5-9ae6-4612-bbac-cad96d3eb3dd | C | Easy |
| 1.29 | de934a98-c3b0-4f0e-bccc-3bbfd3d0308d | B | Easy |
| 1.30 | ab05e5bc-239c-421b-a9ab-056838213249 | C | Easy |
| 1.31 | 2a9be9d6-4a3d-4f2e-92ee-f5423d312669 | A | Easy |
| 1.32 | bd9c2702-bba2-4f95-a9c2-d109cfd98cc1 | B | Easy |
| 1.33 | 077b3b36-9acc-43fd-842b-44f85e95f12f | 30 | Easy |
| 1.34 | 31182667-4cd9-4dea-af99-85f88f84bfa1 | D | Easy |
| 1.35 | d37f3f28-86cf-4511-ab8e-f7639eafd76a | C | Easy |
| 1.36 | bba90b53-4666-4123-babf-8f8dbe2f8a7f | A | Easy |
| 1.37 | af90e68d-767f-4f40-82c5-58dbdc93234b | C | Easy |
| 1.38 | 545e9160-249b-4bdc-a5ec-c8c9e6fc08d6 | D | Easy |
| 1.39 | b057ae02-4dbf-4141-afa7-c0784af1f5b2 | D | Easy |
| 1.40 | 08832-DC | C | Easy |
| 1.41 | 0bd2a2ec-0bbd-4cd3-a37f-8e35808487c2 | B | Easy |
| 1.42 | b1f78aeb-1ccd-4387-9894-38cac3d88e77 | D | Easy |
| 1.43 | a58102d0-5605-4f3b-91b2-44ba46a14ee3 | C | Easy |
| 1.44 | 020395-DC | C | Easy |
| 1.45 | 05103-DC | C | Easy |
| 1.46 | ad28edfc-9193-4278-8a8f-8f9b3a34406c | D | Easy |
| 1.47 | 907da20e-eb78-4bed-85f6-ce91ac870f45 | B | Easy |
| 1.48 | 015564-DC | A | Easy |

| | | | |
|---|---|---|---|
| 1.49 | 025529-DC | B | Easy |
| 1.50 | 0f9a48e3-fac3-492a-8104-05def2d974ce | A | Easy |
| 1.51 | baf1c184-aaa3-416b-ac7d-232d0c6ccc89 | A | Easy |
| 1.52 | a55ac55e-cf83-490d-9ed1-3e835edb8412 | B | Easy |
| 1.53 | 1899a900-ac1b-40ac-a96b-d24684c6d46b | A | Easy |
| 1.54 | 2e557c2f-0543-4d59-933b-131124398b14 | 50 | Easy |
| 1.55 | 08449-DC | D | Easy |
| 1.56 | b63fa866-0dfd-4de5-8017-16574107aa40 | D | Easy |
| 1.57 | 14b71de8-edf5-4dc4-a0fe-e77ab51f082f | C | Easy |
| 1.58 | 7fb9dc2e-7cc3-466f-b4fc-ac67568ce00b | D | Easy |
| 1.59 | 6c1cdc06-5eb3-4cb5-bca2-3f5aa2c8be38 | D | Easy |
| 1.60 | 08177-DC | C | Easy |
| 1.61 | 79c4d360-5538-4b1d-992c-a822445814d8 | A | Easy |
| 1.62 | b4142eb8-7a46-4f1d-a34d-495ec7fb5693 | B | Easy |
| 1.63 | 27ae61df-4f9b-484e-84cb-efeecdfbdc47 | D | Easy |
| 1.64 | ce1f679e-3746-457e-9ae8-995927d4a604 | 11 | Easy |
| 1.65 | 49a43146-5d4a-4013-b46e-e0aa9a32fd1d | C | Easy |
| 1.66 | ec93b358-6d54-4c6a-9e2b-fac8a57da030 | D | Easy |
| 1.67 | a27bd477-cd97-482c-8d5d-ec57f4486fff | D | Easy |
| 1.68 | bfbc2147-51fa-4780-a273-d0cf88011d1c | B | Easy |
| 1.69 | 027789-DC | D | Easy |
| 1.70 | b7b056ed-d7cc-4901-9a1b-f261a1455bee | B | Easy |
| 1.71 | ddfbf736-4c36-47b1-8e69-a9b951ed1842 | D | Easy |
| 1.72 | 72165a25-0c05-4d08-b575-bdacc4a3d7fe | A | Easy |
| 1.73 | fa2330cf-1cd4-47f3-bf5c-5adbf6482253 | B | Easy |
| 1.74 | cdf32e4b-02c1-4140-951c-3af80dec3e9f | D | Easy |

| | | | |
|---|---|---|---|
| 1.75 | b5fb74ca-504a-48fd-8613-f6d71ec02d98 | B | Easy |
| 1.76 | c300a207-fa7b-4c25-876b-37c0579a9bff | C | Easy |
| 1.77 | 017100-DC | C | Easy |
| 1.78 | f2255d61-7225-4d95-b076-0e4c0d8b749d | D | Easy |
| 1.79 | 3f8715bd-7d20-480b-a809-bd3b31a8df26 | B | Easy |
| 1.80 | d3019424-6572-4d26-990d-bc85bde91bda | A | Easy |
| 1.81 | 1a57825e-d5a2-43a1-ad8c-be3b91a796de | C | Easy |
| 1.82 | 0f91f69e-0e9c-452b-aef8-15e6c57f7327 | C | Easy |
| 1.83 | cc329e0c-3593-4615-8425-0a35606bd546 | C | Medium |
| 1.84 | b4b8b378-61b2-456e-8bc3-884fa80f6a40 | 45 | Medium |
| 1.85 | e65207ac-6145-49ce-842e-ade8831b09cf | A | Medium |

## Rationales

▼ 1.1 — 9

The correct answer is $9$. It's given that $gx = 6x$. Substituting $54$ for $gx$ in the given function yields $54 = 6x$. Dividing both sides of this equation by $6$ yields $x = 9$. Therefore, the value of $x$ when $gx = 54$ is $9$.

▼ 1.2 — C

Choice C is correct. It's given that Sam has a flag with an area of $110 \text{ square inches } (\text{in}^2)$ and that each piece of colored nylon Sam sews on increases the flag's area by $64 \text{ in}^2$. If Sam sews on $x$ pieces of nylon and each piece increases the area by $64 \text{ in}^2$, it follows that the expression $64x$ represents the total increase in area, in square inches, from sewing on $x$ pieces of nylon. Adding this increase in area, $64x \text{ in}^2$, to the original area of the flag, $110 \text{ in}^2$, gives the total area of the flag. Therefore, the equation that gives the flag's total area $y$, in square inches, after Sam sews on $x$ pieces of nylon is $y = 110 + 64x$.

Choice A is incorrect. This equation represents the total area, in square inches, of $x$ pieces of colored nylon.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.3 — D

Choice D is correct. It's given that at the beginning of the 1st week of the year there was $\$600$ in a savings account and Gabriella deposits $\$35$ in that savings account at the end of each week. Therefore, the amount of money, in dollars, in the savings account at the end of the 4th week of that year is $600 + 435$, or $740$.

Choice A is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the 4th week if Gabriella withdraws, rather than deposits, $\$35$ at the end of each week.

Choice B is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the 1st week, not the 4th week.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.4 — 2

The correct answer is $2$. Substituting $8$ for $fx$ in the given equation yields $8 = 4x$. Dividing the left- and right-hand sides of this equation by $4$ yields $x = 2$. Therefore, the value of $x$ is $2$ when $fx = 8$.

▼ 1.5 — C

Choice C is correct. For the graph shown, the $x$-axis represents pressure, in atmospheres, and the $y$-axis represents temperature, in kelvins. Therefore, the estimated temperature, in kelvins, of the oxygen gas when its pressure is $6$ atmospheres is represented by the $y$-coordinate of the point on the graph that has an $x$-coordinate of $6$. The point on the graph with an $x$-coordinate of $6$ has a $y$-coordinate of approximately $700$. Therefore, the estimated temperature, in kelvins, of the oxygen gas when its pressure is $6$ atmospheres is $700$.

Choice A is incorrect. This is the pressure, in atmospheres, not the estimated temperature, in kelvins, of the oxygen gas when its pressure is $6$ atmospheres.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.6 — D

Choice D is correct. It's given that the number $y$ is 84 less than the number $x$. A number that's 84 less than the number $x$ is equivalent to 84 subtracted from the number $x$, or $x - 84$. Therefore, the equation $y = x - 84$ represents the relationship between $x$ and $y$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 1.7 — C

Choice C is correct. It's given that this service contract requires a monthly cost of $\$ 23$. A monthly cost of $\$ 23$ for $t$ months results in a cost of $\$ 23t$. It's also given that this service contract requires a onetime activation cost of $\$ 35$. Adding the onetime activation cost to the monthly cost of the service contract for $t$ months yields the total cost $c$, in dollars, of this service contract for $t$ months. Therefore, this situation can be represented by the equation $c = 23t + 35$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.8 — A

Choice A is correct. It's given that Robert was charged an initial fee of $20.00 to rent the truck plus an additional $0.70 per mile driven. Let m represent the number of miles the truck was driven. Since the rental charge is $0.70 per mile driven, $0.70m$ represents the amount Robert was charged for m miles driven. Let c equal the total amount, in dollars, Robert was charged to rent the truck. The total amount can be represented by the equation $c = 20.00 + 0.70m$. It's given that the truck was driven 38 miles, thus $m = 38$. Substituting 38 into the equation gives $c = 20.00 + 0.70(38)$. Multiplying $0.70(38)$ gives $c = 20.00 + 26.60$. Adding these values gives $c = 46.60$, so the total amount Robert was charged is $46.60.

Choices B, C, and D are incorrect and may result from setting up the equation incorrectly or from making calculation errors.

▼ 1.9 — A

Choice A is correct. It's given that the function $g$ is defined by $gx = 4x - 6$. Substituting $-7$ for $x$ into the given equation yields $g\text{-}7 = 4\text{-}7 - 6$, or $g\text{-}7 = -34$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $x$ for which $gx = -7$, not the value of $g\text{-}7$.

▼ 1.10 — 20

The correct answer is $20$. For the graph shown, the $x$-axis represents the time since the initial deposit, in months, and the $y$-axis represents the bank account balance, in dollars. The amount of the initial deposit is estimated by the $y$-coordinate of the point on the graph that represents $0$ months since the initial deposit. Therefore, the amount of the initial deposit is estimated by the corresponding $y$-value for the point when $x = 0$. When $x = 0$, it is estimated that $y = 20$. Thus, the amount of the initial deposit estimated by the graph, to the nearest whole dollar, is $20$.

▼ 1.11 — B

Choice B is correct. The equation for the function f in the xy-plane can be represented by $f(x) = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. Since it's given that the line has a slope of $\frac{3}{4}$, it follows that $m = \frac{3}{4}$ in $f(x) = mx + b$, which yields $y = \frac{3}{4}x + b$. It's given that $f(0) = 3$. This implies that the graph of the function f in the xy-plane passes through the point $(0,3)$. Thus, the y-coordinate of the y-intercept of the graph is 3, so $b = 3$ in $f(x) = \frac{3}{4}x + b$, which yields $f(x) = \frac{3}{4}x + 3$. Therefore, the equation $f(x) = \frac{3}{4}x + 3$ defines the function f.

Choice A is incorrect and may result from a sign error for the y-intercept. Choice C is incorrect and may result from using the denominator of the given slope as m in $f(x) = mx + b$, in addition to a sign error for the y-intercept. Choice D is incorrect and may result from using the denominator of the given slope as m in $f(x) = mx + b$.

▼ 1.12 — B

Choice B is correct. It's given that the equation $d = 16 - \frac{x}{30}$ gives the estimated amount of diesel $d$, in gallons, that remains in the gas tank of the truck after being driven $x$ miles. Substituting $300$ for $x$ in the given equation yields $d = 16 - \frac{300}{30}$, which is equivalent to $d = 16 - 10$, or $d = 6$. Therefore, the estimated amount of diesel that remains in the gas tank of the truck when $x = 300$ is $6$ gallons.

Choice A is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 480$, not when $x = 300$.

Choice C is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 60$, not when $x = 300$.

Choice D is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 0$, not when $x = 300$.

▼ 1.13 — C

Choice C is correct. It's given that the function $f$ is defined by $fx = 25x + 30$. Substituting $2$ for $x$ in this equation yields $f2 = 252 + 30$, which is equivalent to $f2 = 50 + 30$, or $f2 = 80$. Therefore, the value of $fx$ is $80$ when $x = 2$.

Choice A is incorrect. This is the value of $252$, not $252 + 30$.

Choice B is incorrect. This is the value of $25 + 2 + 30$, not $252 + 30$.

Choice D is incorrect. This is the value of $25 + 302$, not $252 + 30$.

▼ 1.14 — B

Choice B is correct. It's given that Josie purchased a pass to tour a nature preserve and $10$ identical stickers for $\$ 62.00$ total. It's also given that the pass cost $\$ 19.50$. Let $x$ represent the cost, in dollars, of one sticker. This relationship can be represented by the equation $19.50 + 10x = 62$. Subtracting $19.50$ from both sides of this equation yields $10x = 42.50$. Dividing both sides of this equation by $10$ yields $x = 4.25$. Therefore, the cost of one sticker was $\$ 4.25$.

Choice A is incorrect. This is the cost of one sticker if Josie purchased $10$ identical stickers for $\$19.50$, not a pass and $10$ identical stickers for $\$62.00$.

Choice C is incorrect. This is the cost of one sticker if Josie purchased $10$ identical stickers for $\$62.00$, not a pass and $10$ identical stickers for $\$62.00$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.15 — A

Choice A is correct. The cost of each additional mile traveled is represented by the slope of the given line. The slope of the line can be calculated by identifying two points on the line and then calculating the ratio of the change in y to the change in x between the two points. Using the points $(1,5)$ and $(2,7)$, the slope is equal to $\frac{7-5}{2-1}$, or 2. Therefore, the cost for each additional mile traveled of the cab ride is $2.00.

Choice B is incorrect and may result from calculating the slope of the line that passes through the points $(5,13)$ and $(0,0)$. However, $(0,0)$ does not lie on the line shown. Choice C is incorrect. This is the y-coordinate of the y-intercept of the graph and represents the flat fee for a cab ride before the charge for any miles traveled is added. Choice D is incorrect. This value represents the total cost of a 1-mile cab ride.

▼ 1.16 — B

Choice B is correct. It's given that the function $f$ is defined by $fx = 3x - 8$. The value of $f7$ is the value of $fx$ when $x = 7$. Substituting $7$ for $x$ in the given equation yields $f7 = 37 - 8$, which is equivalent to $f7 = 21 - 8$, or $f7 = 13$.

Choice A is incorrect. This is the value of $f7$ when $fx = 3x + 8$, rather than $fx = 3x - 8$.

Choice C is incorrect. This is the value of $f1$, rather than $f7$.

Choice D is incorrect. This is the value of $f-7$, rather than $f7$.

▼ 1.17 — 224

The correct answer is $224$. It's given that a student council group uses the function $px = 5x - 220$ to determine their profit $px$, in dollars, for selling $x$ school posters. Substituting $900$ for $px$ in the given function yields $900 = 5x - 220$. Adding $220$ to each side of this

equation yields $1{,}120 = 5x$. Dividing each side of this equation by $5$ yields $224 = x$. Therefore, in order to earn a profit of $\$900$, they must sell $224$ school posters.

▼ 1.18 — 50

The correct answer is $50$. It's given that the function $f$ gives the total number of people on a company retreat with $x$ managers. It's also given that $7$ managers are on the company retreat. Substituting $7$ for $x$ in the given function yields $f7 = 77 + 1$, or $f7 = 50$. Therefore, there are a total of $50$ people on a company retreat with $7$ managers.

▼ 1.19 — A

Choice A is correct. An equation defining a linear function can be written in the form $hx = ax + b$, where $a$ and $b$ are constants. It's given that $h0 = 41$. Substituting $0$ for $x$ and $41$ for $hx$ in the equation $hx = ax + b$ yields $41 = a0 + b$, or $b = 41$. Substituting $41$ for $b$ in the equation $hx = ax + b$ yields $hx = ax + 41$. It's also given that $h1 = 40$. Substituting $1$ for $x$ and $40$ for $hx$ in the equation $hx = ax + 41$ yields $40 = a1 + 41$, or $40 = a + 41$. Subtracting $41$ from the left- and right-hand sides of this equation yields $-1 = a$. Substituting $-1$ for $a$ in the equation $hx = ax + 41$ yields $hx = -1x + 41$, or $hx = -x + 41$.

Choice B is incorrect. Substituting $0$ for $x$ and $41$ for $hx$ in this equation yields $41 = -0$, which isn't a true statement.

Choice C is incorrect. Substituting $0$ for $x$ and $41$ for $hx$ in this equation yields $41 = -410$, or $41 = 0$, which isn't a true statement.

Choice D is incorrect. Substituting $41$ for $hx$ in this equation yields $41 = -41$, which isn't a true statement.

▼ 1.20 — C

Choice C is correct. If $f(x) = \frac{2x-1}{3}$, then $f(5) = \frac{2(5)-1}{3} = \frac{10-1}{3} = \frac{9}{3} = 3$.

Choice A is incorrect and may result from not multiplying x by 2 in the numerator. Choice B is incorrect and may result from dividing 2x by 3 and then subtracting 1. Choice D is incorrect and may result from evaluating only the numerator 2x – 1.

▼ 1.21 — 1

The correct answer is 1. It's given that the function $f$ is defined by $fx = x + \frac{8}{11}$. Substituting $\frac{3}{11}$ for $x$ in the given function yields $f\frac{3}{11} = \frac{3}{11} + \frac{8}{11}$, which gives $f\frac{3}{11} = \frac{11}{11}$, or $f\frac{3}{11} = 1$. Therefore, when $x = \frac{3}{11}$, the value of $fx$ is 1.

▼ 1.22 — 69

The correct answer is 69. The value of $f(20)$ can be found by evaluating the function $f(x) = \frac{7}{10}x + 55$ for $x = 20$. Substituting 20 for $x$ in this function yields $f(20) = \frac{7}{10}(20) + 55$, or $f(20) = 69$. Therefore, the value of $f(20)$ is 69.

▼ 1.23 — B

Choice B is correct. A linear function can be defined by the equation $fx = mx + b$, where $m$ is the slope and $0, b$ is the $y$-intercept of the graph of $y = fx$ in the $xy$-plane. The slope of a line that passes through the points $x_1, y_1$ and $x_2, y_2$ can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $4, 0$ and $0, 8$. Substituting $4, 0$ for $x_1, y_1$ and $0, 8$ for $x_2, y_2$ in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{8 - 0}{0 - 4}$, or $m = -2$. Since the line shown passes through the point $0, 8$, it follows that $b = 8$. Substituting -2 for $m$ and 8 for $b$ in the equation $fx = mx + b$ yields $fx = -2x + 8$. Therefore, the equation $fx = -2x + 8$ defines function $f$.

Choice A is incorrect. This equation defines a linear function whose graph has a slope of -2 and a $y$-intercept of $0, -8$, not $0, 8$.

Choice C is incorrect. This equation defines a linear function whose graph has a slope of 2, not -2, and a $y$-intercept of $0, -8$, not $0, 8$.

Choice D is incorrect. This equation defines a linear function whose graph has a slope of 2, not -2, and a $y$-intercept of $0, 8$.

▼ 1.24 — A

Choice A is correct. It's given that the function $f(x) = 39 - 0.18x$ gives the estimated height, in centimeters (cm), of the fluid in the beaker $x$ days after the start of the experiment. For a function defined by an equation of the form $f(x) = b + mx$, where $m$ and $b$ are constants, $b$ represents the value of $f(x)$ when $x = 0$. It follows that in the given function, 39 represents the value of $f(x)$ when $x = 0$. Since $x = 0$ represents the start of the experiment, then the best interpretation of 39 in this context is that the estimated height, in cm, of the fluid was 39 at the start of the experiment.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. The estimated change in the height, in cm, of the fluid each day is $0.18$, not $39$.

Choice D is incorrect and may result from conceptual errors.

▼ 1.25 — B

Choice B is correct. The *y*-intercept of a graph is the point where the graph intersects the *y*-axis. The graph of function *f* shown intersects the *y*-axis at the point $0, -4$. Therefore, the *y*-intercept of the graph is $0, -4$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.26 — C

Choice C is correct. It's given that *t* represents the number of seconds after the bus passes the marker. Substituting $2$ for *t* in the given equation $d = 30t$ yields $d = 302$, or $d = 60$. Therefore, the bus will be $60$ feet from the marker $2$ seconds after passing it.

Choice A is incorrect. This is the distance, in feet, the bus will be from the marker $1$ second, not $2$ seconds, after passing it.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the distance, in feet, the bus will be from the marker $3$ seconds, not $2$ seconds, after passing it.

▼ 1.27 — A

Choice A is correct. It's given that the function *f* is defined by $fx = 5x + 8$. Substituting $58$ for $fx$ in this equation yields $58 = 5x + 8$. Subtracting $8$ from both sides of this equation yields $50 = 5x$. Dividing both sides of this equation by $5$ yields $10 = x$. Therefore, the value of $x$ when $fx = 58$ is $10$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $f58$, not the value of $x$ when $fx = 58$.

▼ 1.28 — C

Choice C is correct. It's given that $fx$ is the total cost, in dollars, to lease a car from this dealership with a monthly payment of $x$ dollars. Therefore, the total cost, in dollars, to lease the car when the monthly payment is $\$\,400$ is represented by the value of $fx$ when $x = 400$. Substituting $400$ for $x$ in the equation $fx = 36x + 1{,}000$ yields $f400 = 36400 + 1{,}000$, or $f400 = 15{,}400$. Thus, when the monthly payment is $\$\,400$, the total cost to lease a car is $\$\,15{,}400$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.29 — B

Choice B is correct. It's given that the table represents the relationship between the area $x$, in square feet, of the roof of a shopping center and the amount of water $f(x)$, in gallons, drained from the roof into the gutters. Every choice represents this relationship with an equation defining $f$ in the form $f(x) = mx$, where $m$ is a constant rate of change. The value of $m$ can be determined by dividing both sides of the equation by $x$. Each of three pairs of $x$ and $f(x)$ in the table yield $m = 1.8$, since $\frac{4{,}536}{2{,}520} = 1.8$, $\frac{6{,}804}{3{,}780} = 1.8$, and $\frac{9{,}072}{5{,}040} = 1.8$. Therefore, the equation $f(x) = 1.8x$ could define $f$.

Choice A is incorrect. For the roof with an area of $2{,}520$ square feet, this equation would yield $0.6(2{,}520)$, or $1{,}512$, gallons, not the $4{,}536$ gallons shown in the table.

Choice C is incorrect. For the roof with an area of $2{,}520$ square feet, this equation would yield $2{,}268(2{,}520)$, or $5{,}715{,}360$, gallons, not the $4{,}536$ gallons shown in the table.

Choice D is incorrect. For the roof with an area of $2{,}520$ square feet, this equation would yield $4{,}536(2{,}520)$, or $11{,}430{,}720$, gallons, not the $4{,}536$ gallons shown in the table.

▼ 1.30 — C

Choice C is correct. An equation that defines a linear function $f$ can be written in the form $f(x) = mx + b$, where $m$ is the slope of the graph of $y = f(x)$ in the $xy$-plane and $(0, b)$ is the $y$-intercept of the graph. It's given that for the linear function $f$, the graph of $y = f(x)$ in the $xy$-plane has a slope of $7$. Therefore, $m = 7$. It's also given that the graph of $y = f(x)$ in the $xy$-plane passes through the point $(0, 5)$. Therefore, the $y$-intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting $7$ for $m$ and $5$ for $b$ in the equation $f(x) = mx + b$ yields $f(x) = 7x + 5$.

Choice A is incorrect. This equation defines a function whose graph has a slope of $5$, not $7$, and passes through the point $(0, 0)$, not $(0, 5)$.

Choice B is incorrect. This equation defines a function whose graph has a slope of $35$, not $7$, and passes through the point $(0, 0)$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a slope of $12$, not $7$.

▼ 1.31 — A

Choice A is correct. Substituting $63$ for $f(x)$ in the given function yields $63 = 9(2x + 3)$. Dividing both sides of this equation by $9$ yields $7 = 2x + 3$. Subtracting $3$ from both sides of this equation yields $4 = 2x$. Dividing both sides of this equation by $2$ yields $2 = x$. Therefore, $f(x) = 63$ when the value of $x$ is $2$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.32 — B

Choice B is correct. It's given that the function $f$ is defined by $fx = 4x - 3$. Substituting $10$ for $x$ in the given function yields $f10 = 410 - 3$, which is equivalent to $f10 = 40 - 3$, or $f10 = 37$. Therefore, the value of $f10$ is $37$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f10$ for the function $fx = 4x$, not $fx = 4x - 3$.

Choice D is incorrect. This is the value of $f10$ for the function $fx = 4x + 3$, not $fx = 4x - 3$.

▼ 1.33 — 30

The correct answer is 30. The value of $f(x)$ when $x = 4$ can be found by substituting 4 for $x$ in the given equation $f(x) = 7x + 2$. This yields $f(4) = 7(4) + 2$, or $f(4) = 30$. Therefore, when $x = 4$, the value of $f(x)$ is 30.

▼ 1.34 — D

Choice D is correct. It's given that a veterinarian recommends that each day the rabbit should eat 25 calories per pound of the rabbit's weight, plus an additional 11 calories. If the rabbit's weight is $x$ pounds, then multiplying 25 calories per pound by the rabbit's weight, $x$ pounds, yields $25x$ calories. Adding the additional 11 calories that the rabbit should eat each day yields $25x + 11$ calories. It's given that $c$ is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit's weight is $x$ pounds. Therefore, this situation can be represented by the equation $c = 25x + 11$.

Choice A is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat 25 calories per pound of the rabbit's weight.

Choice B is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat $25 + 11$, or 36, calories per pound of the rabbit's weight.

Choice C is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat 11 calories per pound of the rabbit's weight, plus an additional 25 calories.

▼ 1.35 — C

Choice C is correct. The line shown represents the cost $y$, in dollars, for a manufacturer to make $x$ rings. For the line shown, the $x$-axis represents the number of rings made by the manufacturer and the $y$-axis represents the cost, in dollars. Therefore, the cost, in dollars, for the manufacturer to make 60 rings is represented by the $y$-coordinate of the point on the line that has an $x$-coordinate of 60. The point on the line with an $x$-coordinate of 60 has a $y$-coordinate of 175. Therefore, the cost, in dollars, for the manufacturer to make 60 rings is 175.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the cost, in dollars, for the manufacturer to make 20 rings.

Choice D is incorrect. This is the cost, in dollars, for the manufacturer to make 100 rings.

▼ 1.36 — A

Choice A is correct. It's given that $t$ represents the number of monthly deposits. In the given function $ft = 100 + 25t$, the coefficient of $t$ is $25$. This means that for every increase in the value of $t$ by $1$, the value of $ft$ increases by $25$. It follows that with each monthly deposit, the amount in Hana's bank account increased by $\$ 25$.

Choice B is incorrect. Before Hana made any monthly deposits, the amount in her bank account was $\$ 100$.

Choice C is incorrect. After $1$ monthly deposit, the amount in Hana's bank account was $\$ 125$.

Choice D is incorrect and may result from conceptual errors.

▼ 1.37 — C

Choice C is correct. For the graph shown, the $x$-axis represents temperature, in kelvins, and the $y$-axis represents volume, in liters. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is $500$ kelvins is represented by the $y$-coordinate of the point on the graph that has an $x$-coordinate of $500$. The point on the graph with an $x$-coordinate of $500$ has a $y$-coordinate of $7$. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is $500$ kelvins is $7$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.38 — D

Choice D is correct. An equation defining a linear function can be written in the form $fx = mx + b$, where $m$ is the slope and $0, b$ is the $y$-intercept of the graph of $y = fx$ in the $xy$-plane. It's given that the graph of $y = fx$ has a slope of $39$, so $m = 39$. It's also given that the graph of $y = fx$ passes through the point $0, 0$, so $b = 0$. Substituting $39$ for $m$ and $0$ for $b$ in $fx = mx + b$ yields $fx = 39x + 0$, or $fx = 39x$. Thus, the equation that defines $f$ is $fx = 39x$.

Choice A is incorrect. This equation defines a function whose graph has a slope of -$39$, not $39$.

Choice B is incorrect. This equation defines a function whose graph has a slope of $\frac{1}{39}$, not $39$.

Choice C is incorrect. This equation defines a function whose graph has a slope of $1$, not $39$, and passes through the point $0, -39$, not $0, 0$.

▼ 1.39 — D

Choice D is correct. The *y*-intercept of a graph is the point where the graph intersects the *y*-axis. The graph of function *f* shown intersects the *y*-axis at the point $0, 8$. Therefore, the *y*-intercept of the graph of *f* is $(0, 8)$.

Choice A is incorrect. This is the point where the *x*-axis, not the graph of *f*, intersects the *y*-axis.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.40 — C

Choice C is correct. It's given that d is the distance, in miles, Marisol traveled after driving for t hours. Therefore, 45 represents the distance in miles traveled per hour, which is the speed she drove in miles per hour.

Choice A is incorrect and may result from misidentifying speed as the number of trips. Choice B is incorrect and may result from misidentifying speed as the total distance. Choice D is incorrect and may result from misidentifying the speed as the time, in hours.

▼ 1.41 — B

Choice B is correct. It's given that function *f* is defined by $fx = 80 - 6x$. The value of $f7$ can be found by substituting $7$ for *x* in the given function, which yields $f7 = 80 - 67$, or $f7 = 80 - 42$, which is equivalent to $f7 = 38$. Therefore, the value of $f7$ is $38$.

Choice A is incorrect. This is the value of $80 - 67$, not $80 - 67$.

Choice C is incorrect. This is the value of $80 - 61$, not $80 - 67$.

Choice D is incorrect. This is the value of $80 - 6 + 7$, not $80 - 67$.

▼ 1.42 — D

Choice D is correct. It's given that when $x = 0$, $fx = 30$. Substituting $0$ for $x$ and $30$ for $fx$ in the given function yields $30 = 0 + b$, or $30 = b$. Therefore, the value of $b$ is $30$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.43 — C

Choice C is correct. It's given that in the equation $d = 16t$, $d$ represents the distance, in inches, and $t$ represents the number of seconds since an object started moving. In this equation, $t$ is being multiplied by $16$. This means that the object's distance increases by $16$ inches each second. Therefore, the best interpretation of $16$ in this context is that the object is moving at a rate of $16$ inches per second.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This is the best interpretation of $16t$, rather than $16$, in this context.

Choice D is incorrect and may result from conceptual errors.

▼ 1.44 — C

Choice C is correct. It's given that $P(t)$ represents the population of snow leopards t years after 1990. $P(30) = 550$ corresponds to $t = 30$ and $P(t) = 550$. It follows that $t = 30$ corresponds to 30 years after 1990, or 2020. Thus, the best interpretation of $P(30) = 550$ is that the snow leopard population in this area is predicted to be 550 in the year 2020.

Choices A and B are incorrect and may result from reversing the interpretations of t and $P(t)$. Choice D is incorrect and may result from determining that 30 years after 1990 is 2030, not 2020.

▼ 1.45 — C

Choice C is correct. Because f is a linear function of x, the equation $f(x) = mx + b$, where m and b are constants, can be used to define the relationship between x and f ($x$). In this equation, m represents the increase in the value of f (x) for every increase in the value of x

by 1. From the table, it can be determined that the value of f (x) increases by 8 for every increase in the value of x by 2. In other words, for the function $f$ the value of m is $\frac{8}{2}$, or 4.

The value of b can be found by substituting the values of x and f (x) from any row of the table and the value of m into the equation $f(x)=mx+b$ and solving for b. For example, using $x=1$, $f(x)=5$, and $m=4$ yields $5=4(1)+b$. Solving for b yields $b=1$. Therefore, the equation defining the function f can be written in the form $f(x)=4x+1$.

Choices A, B, and D are incorrect. Any equation defining the linear function f must give values of f (x) for corresponding values of x, as shown in each row of the table. According to the table, if $x=3$, $f(x)=13$. However, substituting $x=3$ into the equation given in choice A gives $f(3)=2(3)+3$, or $f(3)=9$, not 13. Similarly, substituting $x=3$ into the equation given in choice B gives $f(3)=3(3)+2$, or $f(3)=11$, not 13.

Lastly, substituting $x=3$ into the equation given in choice D gives $f(3)=5(3)$, or $f(3)=15$, not 13. Therefore, the equations in choices A, B, and D cannot define f.

▼ 1.46 — D

Choice D is correct. Substituting 9 for $x$ in the given equation yields $f9 = 1009 + 2$, or $f9 = 902$. Therefore, the value of $fx$ when $x = 9$ is 902.

Choice A is incorrect. This is the value of $fx$ when $x = 1.09$.

Choice B is incorrect. This is the value of $fx$ when $x = 1.16$.

Choice C is incorrect. This is the value of $fx$ when $x = 8.98$.

▼ 1.47 — B

Choice B is correct. Substituting 50 for $x$ in the given function yields $h50 = 50 + 200$, or $h50 = 250$. Therefore, the value of $h50$ is 250.

Choice A is incorrect. This is the value of $h0$.

Choice C is incorrect. This is the value of $h9{,}800$.

Choice D is incorrect. This is the value of $h50{,}000$.

▼ 1.48 — A

Choice A is correct. It's given that the front of the roller-coaster car starts rising when it's 15 feet above the ground. This initial height of 15 feet can be represented by a constant term, 15, in an equation. Each second, the front of the roller-coaster car rises 8 feet, which can be represented by 8s. Thus, the equation $h = 8s + 15$ gives the height, in feet, of the front of the roller-coaster car s seconds after it starts up the hill.

Choices B and C are incorrect and may result from conceptual errors in creating a linear equation. Choice D is incorrect and may result from switching the rate at which the roller-coaster car rises with its initial height.

▼ 1.49 — B

Choice B is correct. Since Brittany's total take-home pay was $1,576, the value 1,576 can be substituted for T in the given equation $T = 1{,}000 + 18h$ to give $1{,}576 = 1{,}000 + 18h$. Subtracting 1,000 from both sides of this equation gives $576 = 18h$. Dividing both sides of this equation by 18 gives $32 = h$. Therefore, Brittany was paid for 32 hours for her first week of work.

Choice A is incorrect. This is half the number of hours Brittany was paid for. Choice C is incorrect and may result from dividing 1,000 by 18. Choice D is incorrect and may result from dividing 1,576 by 18.

▼ 1.50 — A

Choice A is correct. It's given that the equation $y = 0.67x + 2.6$, where $20 \leq x \leq 110$, gives the predicted number of larvae, $y$, in a colony of ants if the colony has $x$ worker ants. If one of these colonies has $58$ worker ants, the predicted number of larvae in that colony can be found by substituting $58$ for $x$ in the given equation. Substituting $58$ for $x$ in the given equation yields $y = 0.6758 + 2.6$, or $y = 41.46$. Of the given choices, $41$ is closest to the predicted number of larvae in the colony.

Choice B is incorrect. This is closest to the predicted number of larvae in a colony with $87$ worker ants.

Choice C is incorrect. This is closest to the number of worker ants for which the predicted number of larvae in a colony is $58$.

Choice D is incorrect. This is closest to the predicted number of larvae in a colony with $280$ worker ants.

▼ 1.51 — A

Choice A is correct. An equation that defines a linear function $f$ can be written in the form $f(x) = mx + b$, where $m$ is the slope of the graph of $y = f(x)$ in the $xy$-plane and $(0, b)$ is the $y$-intercept of the graph. It's given that for the linear function $f$, the graph of $y = f(x)$ in the $xy$-plane has a slope of $\frac{1}{4}$. Therefore, $m = \frac{1}{4}$. It's also given that the graph of $y = f(x)$ passes through the point $(0, 5)$. Therefore, the $y$-intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting $\frac{1}{4}$ for $m$ and $5$ for $b$ in the equation $f(x) = mx + b$ yields $f(x) = \frac{1}{4}x + 5$.

Choice B is incorrect. This equation defines a function whose graph has a $y$-intercept of $(0, \frac{1}{5})$, not $(0, 5)$.

Choice C is incorrect. This equation defines a function whose graph has a $y$-intercept of $(0, -\frac{5}{4})$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a $y$-intercept of $(0, -5)$, not $(0, 5)$.

▼ 1.52 — B

Choice B is correct. For the graph shown, the $x$-axis represents time, in seconds, and the $y$-axis represents the number of candy bars wrapped. The slope of a line in the $xy$-plane is the change in $y$ for each $1$-unit increase in $x$. It follows that the slope of the graph shown represents the estimated number of candy bars the machine wraps with a label per second. The slope, $m$, of a line in the $xy$-plane can be found using any two points, $x_1, y_1$ and $x_2, y_2$, on the line and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph shown passes through the points $0, 0$ and $2, 80$. Substituting these points for $x_1, y_1$ and $x_2, y_2$, respectively, in the slope formula yields $m = \frac{80 - 0}{2 - 0}$, which is equivalent to $m = \frac{80}{2}$, or $m = 40$. Therefore, the estimated number of candy bars the machine wraps with a label per second is $40$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.53 — A

Choice A is correct. It's given that the function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container $x$ days after the start of the project. For a linear function in the form $f(x) = a + bx$, where $a$ and $b$ are constants, $a$ represents the value of $f(0)$ and $b$ represents the rate of change of the function. It follows that in the given function, $33$ represents the value of $f(0)$. Therefore, the best interpretation of $33$ in this context is the estimated height, in cm, of the liquid at the start of the project.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. The estimated change in the height, in cm, of the liquid each day is $0.18$, not $33$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.54 — 50

The correct answer is $50$. Substituting $8$ for $x$ in the given equation yields $y = 58 + 10$, or $y = 50$. Therefore, the value of $y$ is $50$ when $x = 8$.

▼ 1.55 — D

Choice D is correct. In the xy-plane, the graph of the equation $y = mx + b$, where m and b are constants, is a line with slope m and y-intercept $(0,b)$. Therefore, the graph of $y = 2x - 5$ in the xy-plane is a line with slope 2 and a y-intercept $(0,-5)$. Having a slope of 2 means that for each increase in x by 1, the value of y increases by 2. Only the graph in choice D has a slope of 2 and crosses the y-axis at $(0,-5)$. Therefore, the graph shown in choice D must be the correct answer.

Choices A, B, and C are incorrect. The graph of $y = 2x - 5$ in the xy-plane is a line with slope 2 and a y-intercept at $(0,-5)$. The graph in choice A crosses the y-axis at the point $(0,2.5)$, not $(0,-5)$, and it has a slope of $\frac{1}{2}$, not 2. The graph in choice B crosses the y-axis at $(0,-5)$; however, the slope of this line is $-2$, not 2. The graph in choice C has a slope of 2; however, the graph crosses the y-axis at $(0,5)$, not $(0,-5)$.

▼ 1.56 — D

Choice D is correct. The value of $gx$ when $x = 8$ can be found by substituting $8$ for $x$ in the given equation $gx = 10x + 8$. This yields $g8 = 108 + 8$, or $g8 = 88$. Therefore, when $x = 8$, the value of $gx$ is $88$.

Choice A is incorrect. This is the value of $x$ when $gx = 8$, rather than the value of $gx$ when $x = 8$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.57 — C

Choice C is correct. Each of the tables shows the same three values of $x$: $-1$, $0$, and $1$. Substituting $-1$ for $x$ in the given function yields $g(-1) = 11(-1) + 4$, or $g-1 = -7$. Therefore, when $x = -1$, the corresponding value of $g(x)$ is $-7$. Substituting $0$ for $x$ in the given function yields $g(0) = 11(0) + 4$, or $g(0) = 4$. Therefore, when $x = 0$, the corresponding value of $g(x)$ is $4$. Substituting $1$ for $x$ in the given function yields $g(1) = 11(1) + 4$, or $g(1) = 15$. Therefore, when $x = 1$, the corresponding value of $g(x)$ is $15$. The table in choice C shows $-7$, $4$, and $15$ as the corresponding value of $g(x)$ for $x$-values of $-1$, $0$, and $1$, respectively. Therefore, the table in choice C shows three values of $x$ and their corresponding values of $g(x)$.

Choice A is incorrect. This table shows three values of $x$ and their corresponding values of $g(x)$ for the linear function $g(x) = 4x + 11$.

Choice B is incorrect. This table shows three values of $x$ and their corresponding values of $g(x)$ for the linear function $g(x) = 4x$.

Choice D is incorrect. This table shows three values of $x$ and their corresponding values of $g(x)$ for the linear function $g(x) = 11x$.

▼ 1.58 — D

Choice D is correct. The value of $fx$ when $x = -8$ can be found by substituting $-8$ for $x$ in the given function. This yields $f-8 = -3-8 + 60$, or $f-8 = 84$. Therefore, when $x = -8$, the value of $fx$ is $84$.

Choice A is incorrect. This is the value of $-3 + -8 + 60$, not $-3-8 + 60$.

Choice B is incorrect. This is the value of $-8 + 60$, not $-3-8 + 60$.

Choice C is incorrect. This is the value of $-3 + 60$, not $-3-8 + 60$.

▼ 1.59 — D

Choice D is correct. It's given that Juan rides his bike at an average rate of $5.7$ minutes per mile. The number of minutes it will take Juan to ride $x$ miles can be determined by multiplying his average rate by the number of miles, $x$, which yields $5.7x$. Therefore, the function $mx = 5.7x$ models the number of minutes it will take Juan to ride $x$ miles.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 1.60 — C

Choice C is correct. Substituting 4 for x in g(x) = 5x + a gives g(4) = 5(4) + a. Since g(4) = 31, the equation g(4) = 5(4) + a simplifies to 31 = 20 + a. It follows that a = 11.

Choices A, B, and D are incorrect and may result from arithmetic errors.

▼ 1.61 — A

Choice A is correct. An equation that defines a linear function $f$ can be written in the form $fx = mx + b$, where $m$ and $b$ are constants. It's given in the table that when $x = 0$, $fx = 29$. Substituting $0$ for $x$ and $29$ for $fx$ in the equation $fx = mx + b$ yields $29 = m0 + b$, or $29 = b$. Substituting $29$ for $b$ in the equation $fx = mx + b$ yields $fx = mx + 29$. It's also given in the table that when $x = 1$, $fx = 32$. Substituting $1$ for $x$ and $32$ for $fx$ in the equation $fx = mx + 29$ yields $32 = m1 + 29$, or $32 = m + 29$. Subtracting $29$ from both sides of this equation yields $3 = m$. Substituting $3$ for $m$ in the equation $fx = mx + 29$ yields $fx = 3x + 29$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.62 — B

Choice B is correct. The $y$-intercept of the graph of a function in the $xy$-plane is the point on the graph where $x = 0$. It's given that $f(x) = \frac{1}{10}x - 2$. Substituting $0$ for $x$ in this equation yields $f(0) = \frac{1}{10}(0) - 2$, or $f(0) = -2$. Since it's given that $y = f(x)$, it follows that $y = -2$ when $x = 0$. Therefore, the $y$-intercept of the graph of $y = f(x)$ in the $xy$-plane is $(0, -2)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.63 — D

Choice D is correct. It's given that the function $f(x) = 8x + 4$ gives the estimated height, in feet, of a willow tree $x$ years after its height was first measured. For a function defined by an equation of the form $f(x) = mx + b$, where $m$ and $b$ are constants, $b$ represents the value of $f(x)$ when $x = 0$. It follows that in the given function, $4$ represents the value of $f(x)$ when $x = 0$. Therefore, the best interpretation of $4$ in this context is that the estimated height of the tree was $4$ feet when it was first measured.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 1.64 — 11

The correct answer is $11$. It's given that the function $f(x) = 14 + 4x$ represents the total cost, in dollars, of attending an arcade when $x$ games are played. Substituting $58$ for $f(x)$ in the given equation yields $58 = 14 + 4x$. Subtracting $14$ from each side of this equation yields $44 = 4x$. Dividing each side of this equation by $4$ yields $11 = x$. Therefore, $11$ games can be played for a total cost of $\$58$.

▼ 1.65 — C

Choice C is correct. It's given that the cost of renting a tent is $\$11$ per day for $d$ days. Multiplying the rental cost by the number of days yields $\$11d$, which represents the cost of renting the tent for $d$ days before the insurance is added. Adding the onetime insurance fee

of $\$10$ to the rental cost of $\$11d$ gives the total cost $c$, in dollars, which can be represented by the equation $c = 11d + 10$.

Choice A is incorrect. This equation represents the total cost to rent the tent if the insurance fee was charged every day.

Choice B is incorrect. This equation represents the total cost to rent the tent if the daily fee was $\$d + 11$ for $10$ days.

Choice D is incorrect. This equation represents the total cost to rent the tent if the daily fee was $\$10$ and the onetime fee was $\$11$.

▼ 1.66 — D

Choice D is correct. An equation defining the linear function $f$ can be written in the form $fx = mx + b$, where $m$ is the slope and $0, b$ is the $y$-intercept of the graph of $y = fx$ in the $xy$-plane. It's given that the graph of $y = fx$ has a slope of $2$. Therefore, $m = 2$. It's also given that the graph of $y = fx$ has a $y$-intercept at $0, -5$. Therefore, $b = -5$. Substituting $2$ for $m$ and $-5$ for $b$ in the equation $fx = mx + b$ yields $fx = 2x - 5$. Thus, the equation that defines $f$ is $fx = 2x - 5$.

Choice A is incorrect. For this function, the graph of $y = fx$ in the $xy$-plane has a slope of $\frac{1}{2}$, not $2$.

Choice B is incorrect. For this function, the graph of $y = fx$ in the $xy$-plane has a slope of $-\frac{1}{2}$, not $2$.

Choice C is incorrect. For this function, the graph of $y = fx$ in the $xy$-plane has a slope of $-2$, not $2$.

▼ 1.67 — D

Choice D is correct. It's given that the function $f$ defined by $ft = 14t + 9$ gives the estimated length, in inches, of a vine plant $t$ months after Tavon purchased it. For a function defined by an equation of the form $ft = mt + b$, where $m$ and $b$ are constants, $b$ represents the value of $f0$, or the value of $ft$ when the value of $t$ is $0$. Therefore, for the function defined by $ft = 14t + 9$, $9$ represents the value of $ft$ when the value of $t$ is $0$. This means that $0$ months after the vine plant was purchased, the estimated length of the vine plant was $9$ inches. Therefore, the best interpretation of $9$ in this context is the estimated length of the vine plant was $9$ inches when Tavon purchased it.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. The vine plant is expected to grow $14$ inches, not $9$ inches, each month.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.68 — B

Choice B is correct. Substituting $72$ for $fx$ in the given function yields $72 = 8x$. Dividing each side of this equation by $8$ yields $9 = x$. Therefore, $fx = 72$ when the value of $x$ is $9$.

Choice A is incorrect. This is the value of $x$ for which $fx = 64$, not $fx = 72$.

Choice C is incorrect. This is the value of $x$ for which $fx = 512$, not $fx = 72$.

Choice D is incorrect. This is the value of $x$ for which $fx = 640$, not $fx = 72$.

▼ 1.69 — D

Choice D is correct. The value of $g(0)$ is found by substituting 0 for x in the function g. This yields $g(0) = -0 + 8$, which can be rewritten as $g(0) = 8$.

Choice A is incorrect and may result from misinterpreting the equation as $g(x) = x + (-8)$ instead of $g(x) = -x + 8$. Choice B is incorrect. This is the value of x, not $g(x)$. Choice C is incorrect and may result from calculation errors.

▼ 1.70 — B

Choice B is correct. An equation defining a linear function can be written in the form $fx = mx + b$, where $m$ and $b$ are constants, $m$ is the slope of the graph of $y = fx$ in the $xy$-plane, and $0, b$ is the $y$-intercept of the graph. It's given that for the function $f$, the graph of $y = fx$ in the $xy$-plane has a slope of $3$. Therefore, $m = 3$. It's also given that this graph passes through the point $0, -8$. Therefore, the $y$-intercept of the graph is $0, -8$, and it follows that $b = -8$. Substituting $3$ for $m$ and -8 for $b$ in the equation $fx = mx + b$ yields $fx = 3x - 8$. Thus, the equation that defines $f$ is $fx = 3x - 8$.

Choice A is incorrect. For this function, the graph of $y = fx$ in the $xy$-plane passes through the point $0, 0$, not $0, -8$.

Choice C is incorrect. For this function, the graph of $y = fx$ in the $xy$-plane passes through the point $0, 5$, not $0, -8$.

Choice D is incorrect. For this function, the graph of $y = fx$ in the $xy$-plane passes through the point $0, 11$, not $0, -8$.

▼ 1.71 — D

Choice D is correct. In the given equation, $s$ is the speed, in miles per hour, of a certain car $t$ seconds after it began to accelerate. Therefore, the speed of the car, in miles per hour, $5$ seconds after it began to accelerate can be found by substituting $5$ for $t$ in the given equation, which yields $s = 40 + 35$, or $s = 55$. Thus, the speed of the car $5$ seconds after it began to accelerate is $55$ miles per hour.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.72 — A

Choice A is correct. The $x$-intercept of a graph is the point where the graph intersects the $x$-axis. The graph of function $f$, where $y = fx$, intersects the $x$-axis at $-12, 0$. Therefore, the $x$-intercept of the graph of $f$ is $-12, 0$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.73 — B

Choice B is correct. It's given that the payment $p$, in dollars, for trading in an old computer is the original retail price of the computer minus $\$ 150$ for each year since the computer was released $t$ years ago. It's also given that the original retail price of a computer was $\$ 690$. Thus, the equation that represents this situation is $p = 690 - 150t$.

Choice A is incorrect. This equation represents a situation for a computer with an original retail price of $\$ 150$ minus $\$ 690$ for each year since the computer was released.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.74 — D

Choice D is correct. It's given that the length of the whale was $162 \text{ cm}$ when it was born and that its length increased an average of $4.8 \text{ cm}$ per month for the first $12$ months after it was born. Since $x$ represents the number of months after the whale was born, the total increase in the whale's length, in $\text{cm}$, is $4.8$ times $x$, or $4.8x$. The length of the whale $y$, in $\text{cm}$, can be found by adding the whale's length at birth, $162 \text{ cm}$, to the total increase in length, $4.8x \text{ cm}$. Therefore, the equation that best represents this situation is $y = 4.8x + 162$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 1.75 — B

Choice B is correct. For the graph shown, the $x$-axis represents temperature, in kelvins, and the $y$-axis represents the estimated pressure, in $\text{pounds per square inch psi}$. The estimated pressure of the argon when the temperature is $600$ kelvins can be found by locating the point on the graph where the value of $x$ is equal to $600$. The graph passes through the point $600, 12$. This means that when the temperature is $600$ kelvins, the estimated pressure is $12 \text{ psi}$.

Choice A is incorrect. This is the estimated pressure, in $\text{psi}$, of the argon when the temperature is $300$ kelvins, not $600$ kelvins.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the temperature, in kelvins, of the argon.

▼ 1.76 — C

Choice C is correct. The $y$-intercept of a graph is the point where the graph intersects the $y$-axis. The graph of $y = f(x)$ shown intersects the $y$-axis at the point $(0, 2)$. Therefore, the $y$-intercept of the graph of $y = f(x)$ is $(0, 2)$.

Choice A is incorrect. This is the $x$-intercept, not the $y$-intercept, of the graph of $y = f(x)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.77 — C

Choice C is correct. It's given that the city's minimum hourly wage will increase by $0.50 on the first day of the year for the 5 years after January 1, 2015. Therefore, the total increase, in dollars, in the minimum hourly wage x years after January 1, 2015, is represented by $0.50x$. Since the minimum hourly wage on January 1, 2015, was $9.25, it follows that the minimum hourly wage, in dollars, x years after January 1, 2015, is represented by $9.25+0.50x$. Therefore, the function $f(x)=9.25+0.50x$ best models this situation.

Choices A, B, and D are incorrect. In choice A, the function models a situation where the minimum hourly wage is $9.25 on January 1, 2015, but decreases by $0.50 on the first day of the year for the next 5 years. The functions in choices B and D both model a situation where the minimum hourly wage is increasing by $9.25 on the first day of the year for the 5 years after January 1, 2015.

▼ 1.78 — D

Choice D is correct. It's given that the function $f$ is defined by $fx = \frac{1}{2}x + 6$. Substituting $4$ for $x$ in the given function yields $f4 = \frac{1}{2}4 + 6$, or $f4 = 5$. Therefore, the value of $f4$ is $5$.

Choice A is incorrect. This is the value of $24 + 6$, not $\frac{1}{2}4 + 6$.

Choice B is incorrect. This is the value of $2 + 4 + 6$, not $\frac{1}{2}4 + 6$.

Choice C is incorrect. This is the value of $4 + 6$, not $\frac{1}{2}4 + 6$.

▼ 1.79 — B

Choice B is correct. It's given that the function $f$ is defined by $f(x) = 6.2x$. The value of $f(x)$ when $x = 10$ can be found by substituting $10$ for $x$ in $f(x) = 6.2x$, which yields $f(10) = 6.2(10)$, or $f(10) = 62$. Therefore, when $x = 10$, the value of $f(x)$ is $62$.

Choice A is incorrect. This is the value of $f(10) + 10$, not the value of $f(x)$ when $x = 10$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $6.2 + 10$, not the value of $f(x)$ when $x = 10$.

▼ 1.80 — A

Choice A is correct. The value of $h\text{-}2$ can be found by substituting $-2$ for $x$ in the equation defining $h$. Substituting $-2$ for $x$ in $hx = 3x - 7$ yields $h\text{-}2 = 3\text{-}2 - 7$, or $h\text{-}2 = -13$. Therefore, the value of $h\text{-}2$ is $-13$.

Choice B is incorrect. This is the value of $h\text{-}1$, not $h\text{-}2$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.81 — C

Choice C is correct. An equation defining the linear function $f$ can be written in the form $fx = mx + b$, where $m$ is the slope and $0, b$ is the $y$-intercept of the graph of $y = fx$ in the $xy$-plane. The slope of the graph of $y = fx$ can be found using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$, where $x_1, y_1$ and $x_2, y_2$ are any two points that the graph passes through. If $f0 = 17$, it follows that the graph of $y = fx$ passes through the point $0, 17$. If $f1 = 17$, it follows that the graph of $y = fx$ passes through the point $1, 17$. Substituting $0, 17$ and $1, 17$ for $x_1, y_1$ and $x_2, y_2$, respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{17 - 17}{1 - 0}$, which is equivalent to $m = \frac{0}{1}$, or $m = 0$. Since the graph of $y = fx$ passes through $0, 17$, it follows that $b = 17$. Substituting $0$ for $m$ and $17$ for $b$ in the equation $fx = mx + b$ yields $fx = 0x + 17$, or $fx = 17$. Thus, the equation that defines $f$ is $fx = 17$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.82 — C

Choice C is correct. It's given that the technician charges $\$ 60$ per hour for labor. Therefore, if the job takes $x$ hours, the technician will charge $\frac{\$ 60}{1 \text{ hour}} x \text{ hours}$, or $\$ 60x$, for labor. It's also given that the technician charges $\$ 120$ for parts. Therefore, $fx = 60x + 120$ represents the total amount, in dollars, the technician will charge for this job if it takes $x$ hours.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This function represents the total amount, in dollars, the technician charges for labor only, not the total amount charged for labor and parts.

Choice D is incorrect. This function represents the total amount, in dollars, the technician would charge if the charge for parts were subtracted from, rather than added to, the charge for labor.

▼ 1.83 — C

Choice C is correct. It's given that $f(x) = \frac{x+15}{5}$ and $f(a) = 10$, where $a$ is a constant. Therefore, for the given function $f$, when $x = a$, $f(x) = 10$. Substituting $a$ for $x$ and $10$ for $f(x)$ in the given function $f$ yields $10 = \frac{a+15}{5}$. Multiplying both sides of this equation by $5$ yields $50 = a + 15$. Subtracting $15$ from both sides of this equation yields $35 = a$. Therefore, the value of $a$ is $35$.

Choice A is incorrect. This is the value of $a$ if $f(a) = 4$.

Choice B is incorrect. This is the value of $a$ if $f(a) = 5$.

Choice D is incorrect. This is the value of $a$ if $f(a) = 16$.

▼ 1.84 — 45

The correct answer is $45$. It's given that $h(0) = 45$. Therefore, for the given function $h$, when $x = 0$, $h(x) = 45$. Substituting $0$ for $x$ and $45$ for $h(x)$ in the given function, $h(x) = x + b$, yields $45 = 0 + b$, or $45 = b$. Therefore, the value of $b$ is $45$.

▼ 1.85 — A

Choice A is correct. It's given that the function $g$ models the number of gallons that remain from a full gas tank in a car after driving $m$ miles. In the given function $g(m) = -0.05m + 12.1$, the coefficient of $m$ is $-0.05$. This means that for every increase in the value of $m$ by $1$, the value of $g(m)$ decreases by $0.05$. It follows that for each mile driven, there is a decrease of $0.05$ gallons of gasoline. Therefore, $0.05$ gallons of gasoline are used to drive each mile.

Choice B is incorrect and represents the number of gallons of gasoline in a full gas tank.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.